

Ing. Tomáš Martinovič, Ph.D.

Senior Researcher · IT4Innovations, VSB – Technical University of Ostrava



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Employment History

- 2022-present ■ **Senior Researcher.** IT4Innovations, VSB – Technical University of Ostrava.
- 2019-2022 ■ **Researcher.** IT4Innovations, VSB – Technical University of Ostrava.
- 2016-2018 ■ **Research Assistant.** IT4Innovations, VSB – Technical University of Ostrava.
- 2014-2015 ■ **Junior Data Analyst.** IT4Innovations, VSB – Technical University of Ostrava.

Education

- 2015-2018 ■ **Ph.D., Computational Sciences**, VSB – Technical University of Ostrava.
Thesis title: *Tools for time series analysis of nonlinear dynamical systems.*
- 2012-2015 ■ **Ing., Quantitative Methods in Economy**, Masaryk University.
Thesis title: *Investigating uncertainty in the estimates of potential output and the NAIRU.*
- 2009-2012 ■ **Bc., Mathematical Methods in Economics**, Prague University of Economics and Business.
Thesis title: *Operation research applications in public sector.*

Research Publications

Journal Articles

- 1 Desalegn Chala et al. "Prototype biodiversity digital twin: crop wild relatives genetic resources for food security." *Research Ideas and Outcomes*, 2024. DOI: [10.3897/rio.10.e125192](https://doi.org/10.3897/rio.10.e125192).
- 2 Jürgen Groeneveld et al. "Prototype Biodiversity Digital Twin: honey bees in agricultural landscapes." *Research Ideas and Outcomes*, 2024. DOI: [10.3897/rio.10.e125167](https://doi.org/10.3897/rio.10.e125167).
- 3 Muhammad Iqbal et al. "Demonstration of Sensitive Analysis and Optical Soliton Patterns in a (4+1) Dimensional Boiti-Leon-Manna Pempinelli Equation: Dynamic Insights into Bifurcation, Chaotic Behavior." *Chaos Theory and Applications*, 2025. DOI: [10.51537/chaos.1518307](https://doi.org/10.51537/chaos.1518307).
- 4 Adil Jhangeer et al. "A new technique for characterizing nonlinear dynamical system behavior using self-organizing maps and clustering." *Ain Shams Engineering Journal*, 2026. DOI: [10.1016/j.asej.2026.104206](https://doi.org/10.1016/j.asej.2026.104206).

- 5 Marek Lampart and Tomas Martinovic. "A survey of tools detecting the dynamical properties of one-dimensional families." *Advances in Electrical and Electronic Engineering*, 2017. DOI: [10.15598/aeed.v15i2.2314](https://doi.org/10.15598/aeed.v15i2.2314).
- 6 Stéphane Louise et al. "Modeling and implementing an earthquake and tsunami event-triggered, time-constrained impact assessment workflow." *The International Journal of High Performance Computing Applications*, 2025. DOI: [10.1177/10943420251343794](https://doi.org/10.1177/10943420251343794).
- 7 Tomas Martinovic and Jan Fulneczek. "Fast Algorithm for Contactless Partial Discharge Detection on Remote Gateway Device." *IEEE Transactions on Power Delivery*, 2022. DOI: [10.1109/TPWRD.2021.3104746](https://doi.org/10.1109/TPWRD.2021.3104746).
- 8 T. Martinovič. "Alternative approaches of evaluating the 0-1 test for chaos." *International Journal of Computer Mathematics*, 2020. DOI: [10.1080/00207160.2019.1701662](https://doi.org/10.1080/00207160.2019.1701662).
- 9 Tomáš Martinovič. "Chaotic behaviour of noisy traffic data." *Mathematical Methods in the Applied Sciences*, 2018. DOI: [10.1002/mma.4234](https://doi.org/10.1002/mma.4234).
- 10 Tomáš Martinovič and Georg Zitzlsberger. "Highly scalable algorithm for computation of recurrence quantitative analysis." *The Journal of Supercomputing*, 2019. DOI: [10.1007/s11227-018-2350-5](https://doi.org/10.1007/s11227-018-2350-5).
- 11 Andrii Patrikei et al. "Essential time series characteristics for human motion analysis based on Self-Organizing Map clustering." *Acta Gymnica*, 2026. DOI: [10.5507/ag.2026.004](https://doi.org/10.5507/ag.2026.004).
- 12 Joan Planas-Iglesias et al. "AggreProt: a web server for predicting and engineering aggregation prone regions in proteins." *Nucleic Acids Research*, 2024. DOI: [10.1093/nar/gkae420](https://doi.org/10.1093/nar/gkae420).
- 13 Simon Rolph et al. "Prototype Digital Twin: Recreation and biodiversity cultural ecosystem services." *Research Ideas and Outcomes*, 2024. DOI: [10.3897/rio.10.e125450](https://doi.org/10.3897/rio.10.e125450).
- 14 Allan Souza et al. "Fragmented Networks: Challenges in communication and cohesion of European Biodiversity Research Infrastructures." *Biodiversity Data Journal*, 2025. DOI: [10.3897/BDJ.13.e148079](https://doi.org/10.3897/BDJ.13.e148079).
- 15 Franziska Taubert et al. "Prototype Biodiversity Digital Twin: grassland biodiversity dynamics." *Research Ideas and Outcomes*, 2024. DOI: [10.3897/rio.10.e124168](https://doi.org/10.3897/rio.10.e124168).

Books and Chapters

- 1 Rubén J. García-Hernández et al. "Event-Driven, Time-Constrained Workflows." In *HPC, Big Data, and AI Convergence Towards Exascale*, CRC Press, 2022. DOI: [10.1201/9781003176664-7](https://doi.org/10.1201/9781003176664-7).
- 2 Thierry Goubier et al. "Earthquake and Tsunami Workflow Leveraging the Modern HPC/Cloud Environment in the LEXIS Project." In *Advances in Networked-based Information Systems*, Springer, 2020. DOI: [10.1007/978-3-030-29029-0_21](https://doi.org/10.1007/978-3-030-29029-0_21).
- 3 Thierry Goubier et al. "Real-Time Model of Computation over HPC/Cloud Orchestration - The LEXIS Approach." In *Complex, Intelligent and Software Intensive Systems*, Springer, 2021. DOI: [10.1007/978-3-030-50454-0_24](https://doi.org/10.1007/978-3-030-50454-0_24).
- 4 Marek Lampart and Tomáš Martinovič. "CML-Tent Model Chaotic Behavior with Respect to the State and Coupling Parameters." In *Chaos and Complex Systems*, Springer, 2020. DOI: [10.1007/978-3-030-35441-1_2](https://doi.org/10.1007/978-3-030-35441-1_2).
- 5 Tomáš Martinovič et al. "Automated Application of Inventory Optimization." In *Computer Information Systems and Industrial Management*, Springer, 2016. DOI: [10.1007/978-3-319-45378-1_21](https://doi.org/10.1007/978-3-319-45378-1_21).
- 6 Tomáš Martinovič et al. "Inventory Optimization Model Parameter Search Speed-Up Through Similarity Reduction." In *Soft Computing Applications*, Springer, 2021. DOI: [10.1007/978-3-030-51992-6_9](https://doi.org/10.1007/978-3-030-51992-6_9).
- 7 Tomáš Martinovič et al. "Sequence Analysis for Relationship Pattern Extraction." In *Computer Information Systems and Industrial Management*, Springer, 2019. DOI: [10.1007/978-3-030-28957-7_29](https://doi.org/10.1007/978-3-030-28957-7_29).
- 8 Lukáš Rapant et al. "Traffic Speed Prediction Using Hidden Markov Models for Czech Republic Highways." In *Agent and Multi-Agent Systems: Technology and Applications*, Springer, 2016. DOI: [10.1007/978-3-319-39883-9_15](https://doi.org/10.1007/978-3-319-39883-9_15).

Scientific Projects

Primary investigator

- 2026-2030 ■ **PeatGenie**, project details to be confirmed.
Work Package Leader.
- 2023-2027 ■ **DTO-BioFlow: Integration of biodiversity monitoring data into the Digital Twin Ocean.**
DOI: [10.3030/101112823](https://doi.org/10.3030/101112823). *Task Leader.*
- 2022-2025 ■ **BioDT: Biodiversity Digital Twin for Advanced Modelling, Simulation and Prediction Capabilities.**
DOI: [10.3030/101057437](https://doi.org/10.3030/101057437). *Work Package Leader.*
- 2023-2026 ■ **Apollo data s.r.o. - Application 1 - EyeERP - a comprehensive ERP system for eye clinics with the MojeOči mobile application for performing and analyzing eye tests.**
E122_002/0000436.

International

- 2026-2029 ■ **Czech AI Factory.**
DOI: [10.3030/101314740](https://doi.org/10.3030/101314740). *Work Package Co-Leader.*
- 2025-2028 ■ **LAIFS: LUMI AI Factory - Service Center.**
DOI: [10.3030/101234208](https://doi.org/10.3030/101234208).
- 2025-2027 ■ **InnovAlte Slovakia: Illuminating Pathways for AI-Driven Breakthroughs.**
09I02-03-V01-00029.
- 2023-2026 ■ **EXA4MIND: EXtreme Analytics for MINing Data spaces.**
DOI: [10.3030/101092944](https://doi.org/10.3030/101092944). *Task Leader.*
- 2021-2024 ■ **ACROSS: HPC Big Data Artificial Intelligence Cross Stack Platform Towards Exascale.**
DOI: [10.3030/955648](https://doi.org/10.3030/955648). *Task Leader.*
- 2020-2022 ■ **EuroCC: National Competence Centres in the framework of EuroHPC.**
DOI: [10.3030/951732](https://doi.org/10.3030/951732).
- 2019-2021 ■ **LEXIS: Large-scale EXecution for Industry & Society.**
DOI: [10.3030/825532](https://doi.org/10.3030/825532). *Task Leader.*

National

- 2026-present ■ **Národní Centrum Umělé Inteligence**, details to be confirmed.
- 2021-2023 ■ **SmartFleet - AI-based software for full use of electric vehicles in companies and maximizing their share in the vehicle fleet.**
EG20_321/0024896.
- 2019-2021 ■ **Contact-less detector for partial discharges activity in medium overhead voltage powerlines.**
TJ02000031.
- 2016-2020 ■ **IT4Innovations excellence in science.**
LQ1602.

Contractual research

2016-2017  **IXPERTA s.r.o.**

2015-2016  **K2 Atmitec s.r.o.**

Teaching Experience

University Courses

2025/2026  **Libraries for parallel data processing.**
Summer

Tutor, teacher giving lectures.

2025/2026  **Libraries for parallel data processing.**
Winter

Tutor, teacher giving lectures.

2024/2025  **Libraries for parallel data processing.**
Winter

Tutor, teacher giving lectures.

2023/2024  **Libraries for parallel data processing.**
Winter

Tutor, teacher giving lectures.

2022/2023  **Libraries for parallel data processing.**
Winter

Tutor, teacher giving lectures.

2021/2022  **Libraries for parallel data processing.**
Winter

Tutor, teacher giving lectures.

2020/2021  **Libraries for parallel data processing.**
Winter

Tutor, teacher giving lectures.

2019/2020  **Libraries for parallel data processing.**
Winter

Tutor, teacher giving lectures.

Bachelor/Diploma Thesis

2026  Petr Kladov, *Kolmogorov-Arnold Network analysis and comparison with classical neural networks*, VSB - Technical University of Ostrava.

Diploma Thesis - Supervisor.

2023  Matěj Eliáš, *Tool for iterative visualization of selected protein characteristics in a web interface*, VSB - Technical University of Ostrava.

Bachelor Thesis - Supervisor.

2020  Jan Pastorek, *Use of High-level Synthesis to accelerate application development to programmable logic*, VSB - Technical University of Ostrava.

Bachelor Thesis - Consultant.

Training Events

May 2026  AI Confidential, Prague.

April 2026  AI Confidential, Brno.

April 2026  AI Confidential, Ostrava.

June 2025  Python for Data Stewards, Ostrava.

April 2025  Python for HPC, Online.

May 2024  Parallel R, Online.

April 2023  High Performance Data Analysis with R, Ostrava/Hybrid.

May 2022  Data science with R and Python, Online.

September 2020  Applied Image Segmentation using Intel Architecture, Online.

- July 2020  Deep Learning on Intel Architecture, Online.
- May 2020  Deep Learning on Intel Architecture, Online.
- July 2019  Intel AIDC Summit, Munich.
- June 2019  Hands-on Developer Workshop for Artificial Intelligence, Milan.
- April 2019  Data science with R and Python - Ostrava.

Summer School Lectures

- March 2025  **Organising committee and lectures**, BioDT School, Lecce.
- August 2024  **Data Science with R**, EUMaster4HPC, Summer School, Ostrava.
- June 2023  **Data Science with R**, SCTrain Summer School, Ostrava.

Miscellaneous Experience

Awards

- June 2019  **Best Poster Award**, *HPC Oriented Algorithm for Computation of Recurrence Quantitative Analysis*, ISC High Performance 2019.
- May 2019  **1st Place in The Competition for the Best Dissertation Defended in 2018, Topic: Information Technology and Electrical Engineering.**
Tools for time series analysis of nonlinear dynamical systems, VSB - Technical University of Ostrava and Consortium PROGRES 3.

Scientific Reviews

- since 2026  Frontiers in Plant Science.
- since 2026  SKV - Systém kvalitních výsledků.
- since 2025  Journal of Science and Technology.
- since 2024  Ecological Informatics.
- since 2024  Ecological Modelling.
- since 2019  International Journal of Computer Mathematics.
- since 2018  The Journal of Supercomputing.
- since 2018  Mathematical Methods in the Applied Sciences.
- since 2016  CISIM - International Conference on Computer Information Systems and Industrial Management.

Participation in Organisations and Conference

- since 2024  ETP4HPC SRA7 HPC / AI Applications and Workflows.
- since 2020  BDVA Task Force Finance.
- 2026  TDWG 2026 co-convening session on *Digital Twins, Data Infrastructures*.
- since 2026  ELIXIR CZ AI Positioning framework.

International Mobility

- June-November 2018  Politecnico di Milano, Dipartimento di Elettronica, Informazione e Bioingegneria.

Other

- June 2024  **Invited Panel Discussion.** Distributed Digital Twins: Engineering Challenges and Future Trends.